DCiROM: A High-Density Fully-Digital Compute-in-Read-Only-Memory Macro for Energy-Efficient Task-Level DNN Inference

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***Abstract*—Compute-in-memory (CiM) is a promising approach to alleviating the von Neumann memory wall bottleneck in data-intensive tasks. As a candidate to build CiM, static-random-access memory (SRAM) has become popular in accelerating deep neural networks (DNNs) because of its mature fabrication support and high flexibility. However, SRAM-based CiM suffers from low memory density, resulting in frequent weight reloading and increased power consumption during end-to-end inference tasks. Recently, read-only-memory (ROM)-based analog CiM (ACiM) has demonstrated the potential to deploy all DNN weights on-chip by leveraging high-density ROM. It may also achieve encouraging flexibility by exploiting the synergy between a small SRAM and a big ROM. Nevertheless, it faces challenges in achieving high computing density due to limitations in analog-to-digital converter (ADC) performance. Meanwhile, digital CiM (DCiM) improves computing density but incurs significant energy and area overhead due to the adder tree. To address these challenges, this paper proposes the first fully digital and flexible compute-in-read-only-memory (DCiROM) design approach. On the one hand, DCiROM introduces a novel ROM-logic fusion CiM that reduces the CiM area by 51% while maintaining high memory density and computing performance. On the other hand, DCiROM achieves high flexibility with minimal area overhead by proposing two innovative multiply-and-accumulate (MAC) resource-reusing techniques. This work has implemented a DCiROM chip with 3024Kb ResNet-56 parameters using a 65nm CMOS technology. This macro achieves 6.9-29.5x higher normalized FoM (memory density x computing density) than the state-of-the-art CiM works. Additionally, it reduces energy consumption per image inference by 2.9-9.9x compared to SRAM-based CiM works when off-chip access is considered.**

***Index Terms*—Compute-in-memory (CiM), read-only memory (ROM), multiply-and-accumulation (MAC), deep neural network (DNN).**

I. Introduction

DEEP neural networks (DNNs) have found widespread application across various machine learning domains,including computer vision, natural language processing, and autonomous driving [1], [2], [3]. However, the exponentially increasing volume of data and computational demands exacerbate the "memory wall bottleneck" inherent in the von Neumann architecture [4], [5]. This persistent bottleneck arises from the significant mismatch in time and power consumption between processing unit (PU) computation and memory-PU data transfer. Consequently, the performance of AI processors is severely constrained by the frequent data movement required for DNN processing.

To address this challenge, compute-in-memory (CiM) [6], [7] has emerged as a promising approach to mitigating the frequent data movement between external memory and PUs. Among the CiM technologies currently being explored, SRAM-based CiM stands out because of its remarkable performance enhancements in handling data-intensive tasks on AI devices [8], [9], [10], [11].

Although SRAM CiM has high flexibility and mature fabrication support, its low memory density caused by large cell area makes it challenging to store all the weights of DNNs on-chip. Current SRAM CiM works typically employ 6T [9], [11], 8T [12], [13], or 10T [14] basic cells, which occupy a significant silicon area. Some works utilize foundry-provided compact-layout 6T SRAM cells and local computing cells (LCCs) [9], [15]. However, the on-chip capacity remains limited due to low memory density. Consequently, SRAM CiM accelerators require loading weights from off-chip DRAM during end-to-end inference tasks, resulting in undesirable energy consumption and latency. To address these on-chip capacity limitations, compute-in-read-only-memory (ROM) has been proposed as a viable solution [16], [17], [18], [19]. The high-density analog ROM-based CiM provides sufficient storage capacity to store the entire DNN on-chip, thereby reducing the power and delay associated with off-chip data transfers, as shown in Fig. 1.

However, ROM CiM faces challenges in flexibility due to its fixed weights. Efforts to address this issue include YOLoC [20] and Hidden-ROM [21], which have made significant progress in overcoming the flexibility limitation of ROM CiM. These methods propose hybrid architectures that combine low-capacity SRAM CiM and high-capacity ROM CiM at cell-level and array-level, enabling adaptation to various DNN inference tasks. Nevertheless, there remain additional opportunities for optimization in the design of the multiply-and-accumulate (MAC) circuits within CiM architecture.

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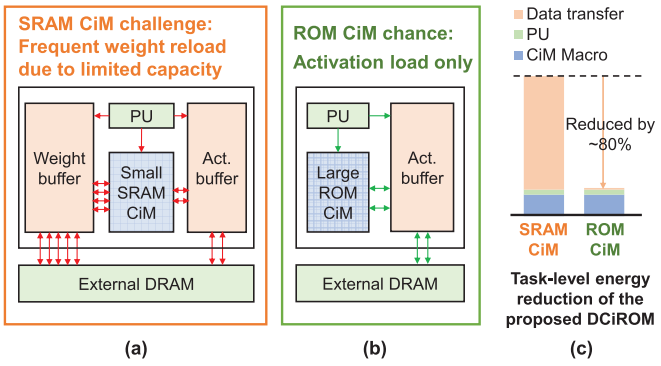


Fig. 1. The significance of high-density CiM: (a) Challenges of SRAM-based CiM, (b) Opportunities of ROM-based CiM, (c) Transport energy reduction in ROM-based CiM.

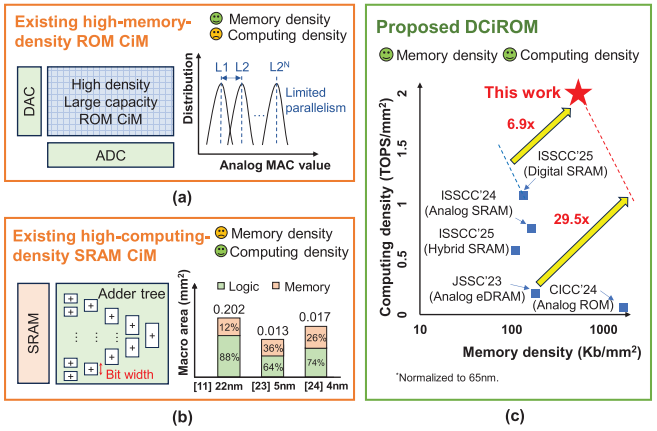


Fig. 2. Conflict between (a) memory density and (b) computing density, and (c) highlights of the proposed DCiROM.

Additionally, existing analog ROM CiM typically suffers from low throughput and limited computing parallelism due to analog-to-digital converter (ADC) [22]. On the other hand, digital CiM [11], [23], [24] generally achieves better computing parallelism but is limited by lower memory density, primarily due to the significant area overhead of adder trees, as illustrated in Fig. 2.

This work proposes a fully digital compute-in-ROM design approach to address the aforementioned challenges. The key contributions of this work include:

- DCiROM: A novel ROM-logic fusion design approach is proposed, achieving high memory density and high computing density through a significantly simplified ROM circuit and adder tree.
- Flexibility: A DCiROM flexibility architecture, based on YOLoC, is implemented. Two ROM CiM MAC resource reuse methods are introduced to further reduce computational overhead while enabling seamless task migration.
- Verification: A 65nm DCiROM chip has been designed and fabricated, incorporating all weights of ResNet-56. The chip achieves an experimentally measured ultra-high computing density of 2.06 TOPS/mm^2 and a memory density of 487 Kb/mm^2 for end-to-end inference tasks.

In the remainder of this paper, Section II reviews the background of analog ROM CiM and ROM CiM flexibility. Section III presents the proposed chip architecture and optimization methods. Section IV shows the measured performance results with a fabricated chip. Section V discusses the overhead and future work. Section VI concludes this work.

II. Background

This section briefly overviews existing ROM-based Compute-in-Memory (CiM) implementations within the analog domain. Additionally, it discusses the challenges related to flexibility in ROM-based CiM architectures and introduces YOLoC, a recent advancement that significantly improves the flexibility of ROM CiM systems.

A. Analog ROM CiM

To improve the on-chip capacity of CiM, non-volatile memory technologies have been considered, including RRAM [25], MRAM [26], and PCRAM [27]. However, these technologies still face challenges such as immature fabrication processes, limited endurance, and high write energy. An alternative approach is using embedded DRAM (eDRAM) [22], [28], [29]. eDRAM offers a smaller cell size compared to SRAM while retaining rewritability. Nevertheless, eDRAM requires frequent refresh operations to address charge leakage, resulting in higher energy consumption.

Read-only memory (ROM) is a practical solution for enhancing on-chip capacity due to its high storage density and reliability. Existing ROM-based CiM works utilize 1T ROM cells [16], [17], [18], [19]. Data are stored by connecting the transistor and different bitlines (BLs). The multiplication operation is realized by using the inherent switching characteristics of the 1T cells. Computing modes include charge domain [16], [17], [19], and current domain [18]. Additionally, 2-bit multi-level cell (MLC) ROM cells [16], [17], [18] have been utilized, achieving an on-chip storage capacity of 22 Mb on a single macro [17]. Meanwhile, [19] utilizes quad-level cell (QLC) ROM cells, achieving an on-chip memory density of 19.7 Mb/mm^2. However, due to signal margin limitations in ADCs, analog ROM CiM designs typically operate with low computing parallelism to ensure inference accuracy, which ultimately restricts computing density.

B. ROM CiM Flexibility and YOLoC

While ROM CiM achieves high energy efficiency at task level, achieving flexibility remains challenging due to its fixed weights, as illustrated in Fig. 3(a). Although tasks with stationary algorithms and scenarios, such as optical character recognition (OCR) and biometric recognition, can be directly deployed on ROM CiM, they cannot support transitions between different tasks. To address this limitation, YOLoC [20] introduced a novel architecture called "ReBranch," which enables seamless transfer of ROM CiM. Inspired by the residual block of ResNet, ReBranch consists of two parallel blocks: a trunk for fixed convolution using ROM CiM and a branch for residual convolution using SRAM CiM, as shown

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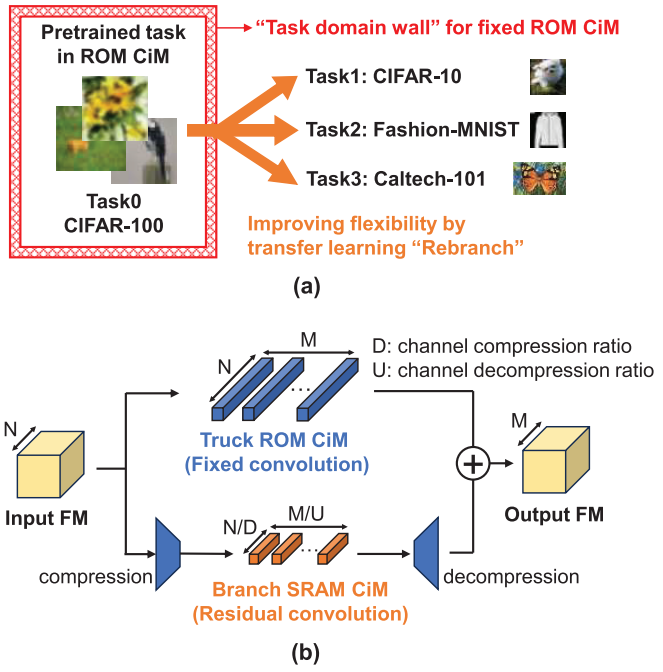


Fig. 3. "Task domain wall" in ROM CiM: (a) Flexibility challenges in ROM CiM, (b) "ReBranch" architecture of YOLoC [20], which combines truck ROM CiM and branch SRAM CiM to achieve flexibility.

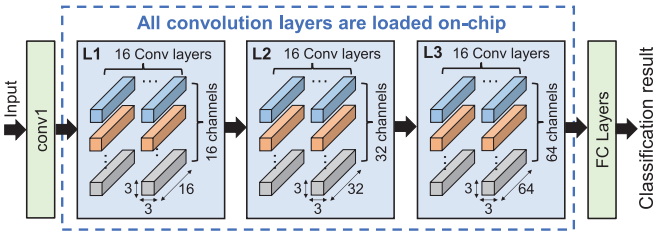


Fig. 4. Structure of ResNet-56 [30].

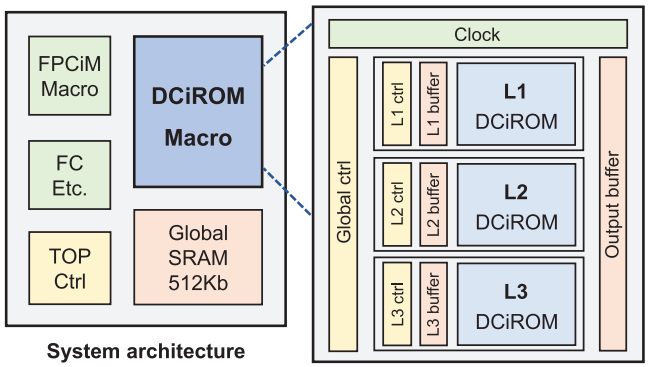


Fig. 5. System and macro structures.

in Fig. 3(b). Compression and decompression layers by ROM CiM are added before and after the branch to decrease the parameters and computations. As a result, the branch can be realized using a small-scale SRAM CiM. By training only the branch weights on different datasets, ROM CiM can achieve a certain degree of flexibility.

ReBranch has demonstrated robust task transition flexibility across various computer vision tasks, such as transitioning from CIFAR-100 to Caltech-101 [16]. Previous work [17] implemented ReBranch architecture to support low-rank adaptation (LoRA) in transformer-based large-language models (LLMs) [31], proving the effectiveness of ReBranch in achieving ROM CiM flexibility. The truck and branch are implemented independently using analog-domain ROM CiM and SRAM CiM. Despite their different array structures, this work notices that both ROM CiM and SRAM CiM utilize the same MAC circuits (ADCs or adder trees). This suggests that reusing these circuits represents a promising optimization direction that has yet to be explored.

III. Proposed Digital ROM CiM Macro

This work proposes the first fully digital ROM CiM macro for multi-bit MAC operations. The proposed macro utilizes three digital compute-in-ROM (DCiROM) blocks to further enhance computing density compared with previous analog ROM CiM macros. This section details the structure and operations of the proposed macro, followed by a discussion on ROM CiM reuse for achieving task-level flexibility with minimal area overhead.

A. Structure of DCiROM Macro

This chip loads all convolution layers of ResNet-56 (excluding nonlinear layers). Fig. 4 provides a detailed breakdown of the ResNet-56 structure. The convolution layers are divided into three main sets, labeled L1, L2, and L3. Each set comprises 16 convolution layers with the following dimensions: height = 3/3/3, width = 3/3/3, depth = 16/32/64, and channel number = 16/32/64, respectively. The parameters loaded in L1, L2, and L3 are statically quantized to 4-bit values in 2's complement format.

Fig. 5 illustrates the system architecture of ResNet-56. The system consists of a DCiROM macro, a floating-point (FP) CiM macro, a fully connected (FC) processor, a top controller, and a 512Kb global SRAM. The FPCiM macro serves as the feature extractor for the "conv1" shown in Fig. 4. The DCiROM macro is fabricated as a chip using a 65nm CMOS technology. The chip comprises three DCiROM cores for L1, L2, and L3, along with a clock module, controllers, and input/output buffers.

Fig. 6 shows the block-level architecture of the proposed DCiROM macro. Each DCiROM core contains multiple DCiROM slices. Within each slice, 16 ROM columns share a single digital MAC unit to achieve high memory density, enabling all 3024Kb parameters to be stored on-chip. During a MAC operation, 4-bit activations are transmitted to the DCiROM cores using a bit-serial approach, while the 4-bit weights are stored across 4 adjacent DCiROM slices. In each cycle, the sums from the four adjacent DCiROM slices are shifted and accumulated to compute the weight partial sum. After 4 cycles, the weight partial sums corresponding to the 4 activation bits are shifted and added to produce

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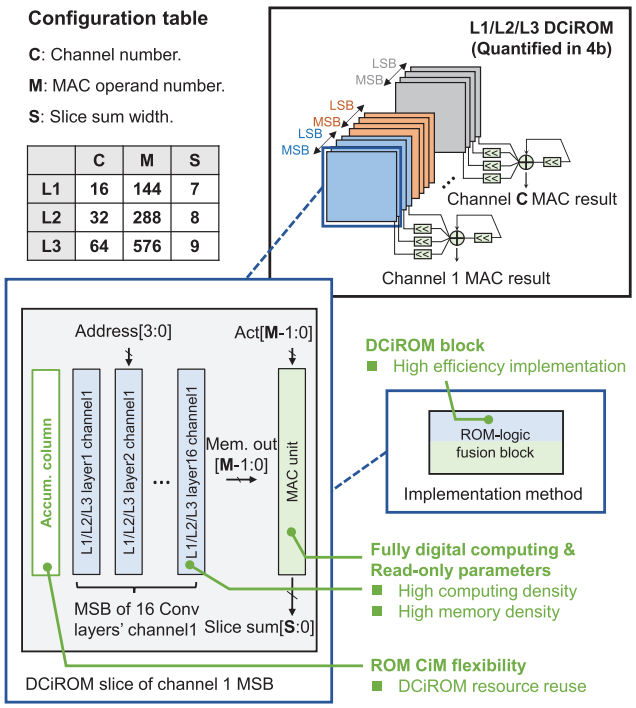


Fig. 6. Architecture of the DCiROM macro.

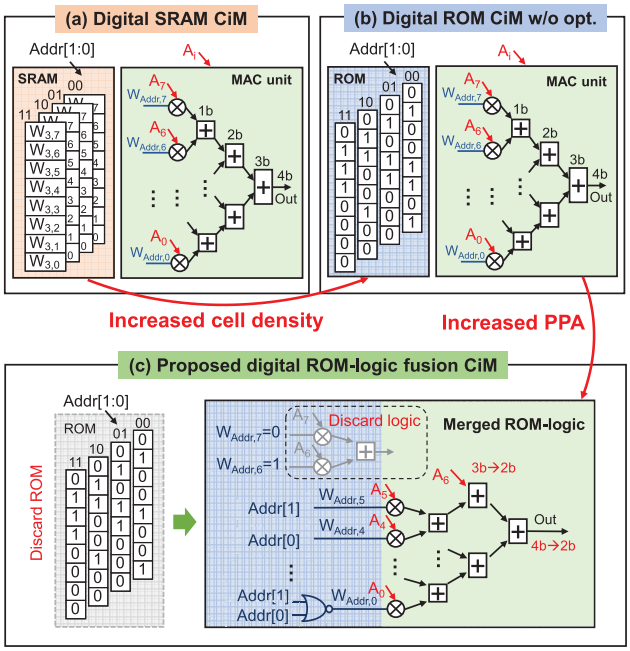


Fig. 7. Density enhancement process: (a) Digital SRAM CiM, (b) Digital ROM CiM without optimization, (c) Proposed digital ROM-logic fusion CiM (illustrating an example of 8 1-bit activations and 8 1-bit weights MAC operations in a 4-column memory).

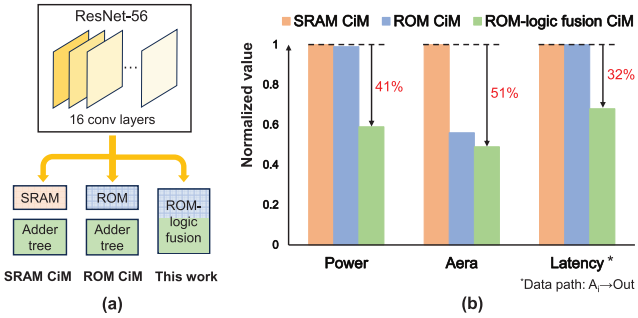


Fig. 8. Evaluation of the ROM-logic fusion CiM: (a) Implementation of ResNet-56 convolution layers, (b) PPA comparison of the three methods.

the final MAC output. The DCiROM slices are implemented using ROM-logic fusion blocks, which innovatively utilize fixed parameters for optimization. Additionally, an original ROM CiM resource reuse method is employed to reduce area overhead while achieving computing flexibility.

B. Synthesizable ROM-Logic Fusion CiM

Fig. 7 illustrates the logic-level process of enhancing memory density from digital SRAM CiM to the proposed ROM-logic fusion CiM. A conventional digital SRAM CiM macro comprises a memory block and an adder tree, as shown in Fig. 7(a). Generally, the adder tree occupies more than 64% of the macro area to ensure computing parallelism [11], [23], [24]. To improve on-chip memory density, the SRAM array can be replaced with a ROM array, getting the digital ROM CiM without optimization, as shown in Fig. 7(b).

Furthermore, the memory and MAC logic can be simplified because of the fixed weights stored in ROM. This optimization can be divided into two parts: memory and MAC unit, ultimately resulting in a ROM-logic fusion block. Considering the 4-bit weights in ResNet-56, each data point typically requires 4 SRAM cells. However, the bit-width of a ROM block can be reduced if one readout bit remains unchanged. For the MAC unit, fixed "1" and "0" values can be utilized to simplify the computing logic, as shown in Fig. 7(c). Given the sparsity of DNN parameters, a significant number of "0" bits exist in the weights, allowing the corresponding multipliers and adders to be eliminated. This approach facilitates the precise optimization of adder tree while effectively mitigating soft errors inherent in conventional array-based memory architectures. The optimization process can be efficiently carried out by electronic design automation (EDA) tools, such as Synopsys Design Compiler. As a result, the ROM-logic fusion CiM represents a synthesizable design approach.

Fig. 8 illustrates the optimization effects of ROM-logic fusion CiM. As shown in Fig. 8(a), the 16 convolution layers of L1 are loaded onto three platforms: conventional digital SRAM CiM, ROM CiM without optimization, and the proposed ROM-logic fusion CiM. Fig. 8(b) compares the power, performance, and area (PPA) of the three methods. For the comparison, the memory cell in push-rule is utilized, and the peripheral overhead such as decoders and sense-amplifiers are excluded. The proposed ROM-logic fusion block achieves a 41% reduction in power, a 51% reduction in area, and a 32% reduction in latency compared to conventional digital SRAM CiM. Additionally, it reduces area by 44% compared to digital ROM CiM without optimization.

The PPA improvement of ROM-logic fusion CiM is influenced by data sparsity and the memory-compute ratio. Fig. 9

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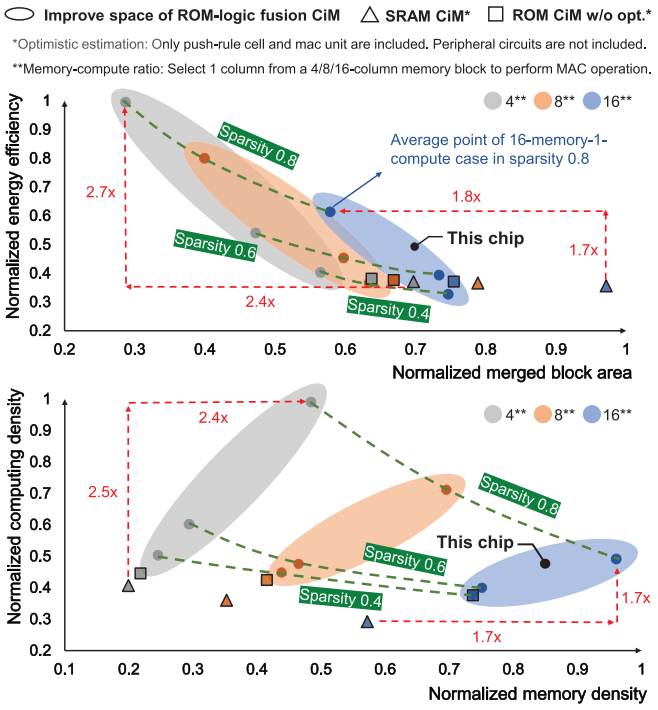


Fig. 9. Impact of memory-compute ratio and data sparsity on ROM-logic fusion CiM optimization.

shows the trade-offs between memory density, computing density, and energy efficiency of the proposed ROM-logic fusion CiM. Generally, higher data sparsity leads to better PPA. A larger memory-compute ratio results in higher memory density, but lower energy efficiency and computing density. The extent of optimization varies across different network layers depending on their data sparsity, forming an elliptical range when plotted. The square and triangle markers represent conventional digital SRAM CiM and digital ROM CiM without optimization, respectively. Markers of different colors indicate varying memory-compute ratios. Gate-level simulation results demonstrate that ROM-logic fusion CiM achieves 1.7-2.4x higher memory density, 1.7-2.5x higher computing density, and 1.7-2.7x higher energy efficiency in sparse patterns compared to digital SRAM CiM.

This chip adopts a memory-compute ratio of 16 to store all 3024Kb parameters on-chip. The data sparsity of L1, L2, and L3 is 0.71, 0.69, and 0.65, respectively. It is important to note that if the memory-compute ratio becomes excessively large, the merged adder tree in the ROM-logic fusion CiM will degenerate to a generic adder tree, and the logic memory will become highly complicated. In such cases, the ROM-logic fusion CiM may not outperform the digital ROM CiM without optimization.

$$\begin{aligned} BRANCH_OUT &= \sum_n A_n W_n \\ &= \sum_i \sum_j 2^{i+j} \sum_n A_{n,i} W_{n,j} \\ &= \sum_i \sum_j 2^{i+j} \sum_n Prod_{n,i,j} \end{aligned} \tag{1}$$

TABLE I
REBRANCH CONFIGURATION

	Truck			Branch					
Layer	L1	L2	L3	L1		L2		L3	
D x U*	-			4	16	4	16	4	16
Operand**	144	288	576	72	36	144	72	288	144
Accumulator	16	32	64	8	4	16	8	32	16
#ROM (Kb)	9	36	144	1	0.5	4	2	16	8
#SRAM (Kb)	-			2.25	0.56	9	2.25	36	9

*D: compression ratio, U: decompression ratio [20], in this work D = U.

**The number of MAC inputs, the operand of a 3x3x16 convolution kernel is 144.

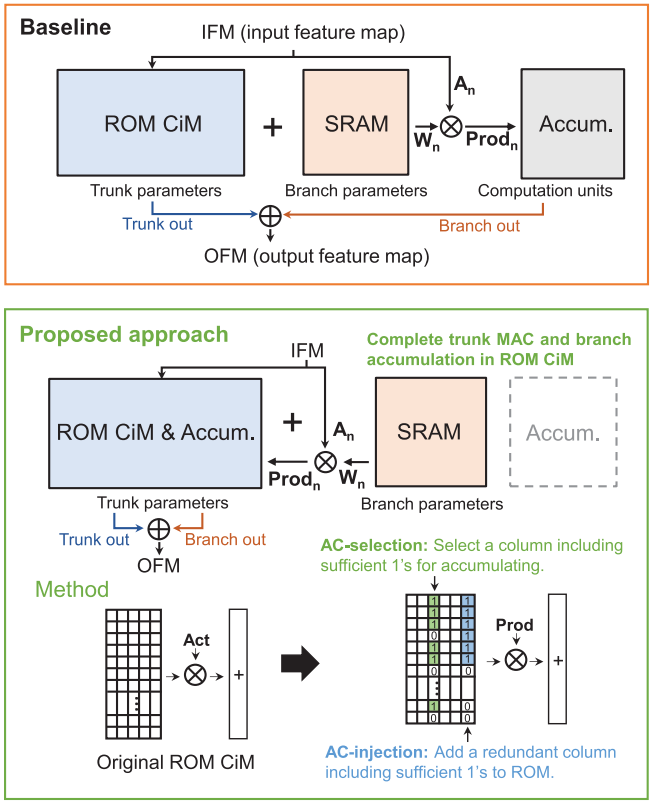


Fig. 10. Methods to achieve flexibility through ROM CiM resource reuse: AC-selection and AC-injection.

C. ROM CiM Resource Reuse

This work employs the ReBranch architecture, as described in Section II-B, to achieve flexibility in ROM CiM. Table I presents the ReBranch configuration of the ROM CiM truck and the SRAM CiM branch. Generally, a higher DxU (Downsample and Upsample) requires fewer branch computing resources but may result in greater inference accuracy loss. In this chip, the DxU is set to 16.

Fig. 10 illustrates the ROM CiM resource reuse mechanism. The branch computation is the MAC operation of the input feature map (IFM) and trainable parameters. This MAC operation can be decomposed into 1-bit by 1-bit multiplication and accumulation. As a result, additional SRAM, multipliers,

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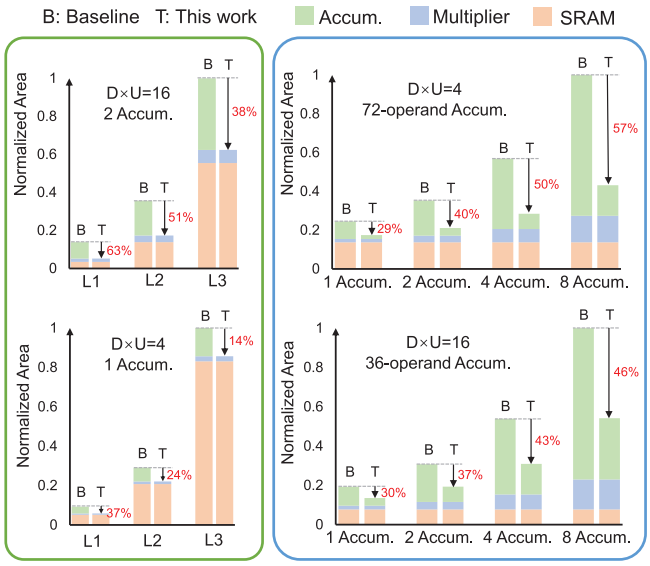


Fig. 11. Simulation results of ROM CiM resource reuse.

and accumulators are required to support the branch MAC computation. In previous analog ROM CiM works [16], [17], the branch circuit is implemented as a separate SRAM CiM block. However, in the digital domain, ROM CiM resources can be reused to accumulate partial products. The underlying theory is presented in Equation 1. Each DCiROM channel, as shown in Fig. 6, can be repurposed as an accumulator for the branch SRAM CiM.

Given that the ROM-logic CiM is simplified by fixed weights, it cannot directly compute branch MAC operations. To address this, there must be a sufficient number of 1's in the columns of a DCiROM channel to satisfy the branch operands. This work proposes two methods, as shown in Fig. 10: accumulator column (AC)-selection and AC-injection. AC-selection (ACSEL) selects columns with enough 1's from the original ROM slice, while AC-injection (ACINJ) adds a redundant column with enough 1's to the ROM slice. Generally, ACSEL reduces area overhead more effectively, but ACINJ is better suited for very sparse applications.

Simulation results of the proposed two methods are presented in Fig. 11. In the case of ACSEL, L1/L2/L3 can be repurposed to function as two 36/72/144-operand accumulators when the compression ratio is 16. The branch area overhead is reduced by 63%/51%/38% compared with the corresponding baseline. For ACINJ, the branch MAC parallelism depends on the number of columns injected. The branch area overhead is reduced by 29-57% compared to the baseline. In this chip, L1/L2/L3 can be reused to perform as four 36/72/144-operand accumulators using ACINJ, achieving a 43% reduction in branch area compared with the corresponding baseline.

In ResNet-56, the arithmetic requirements of deeper layers (L2/L3) are significantly reduced compared to earlier layers (L1) due to progressive feature map (IFM) downscaling. Measurements indicate that L2 and L3 exhibit 3x and 8x lower execution latency, respectively, during end-to-end inference.

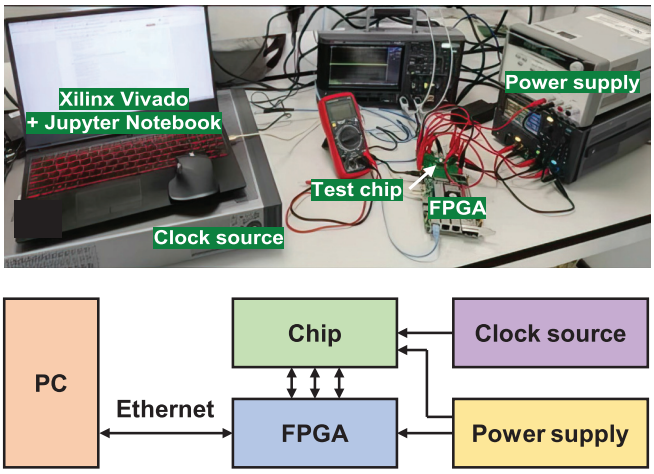


Fig. 12. Test platform of the fabricated chip.

To optimize hardware utilization, L2/L3 compute resources are time-division multiplexed (TDM) for branch MAC operations. Consequently, the proposed ROM CiM resource reuse mechanism introduces minimal throughput overhead, with only 9% area and 5% energy penalties.

IV. Measurement Results

This section presents the performance evaluation and experimental results of the proposed schemes. Owing to its significantly higher density, the proposed design exhibits a remarkably lower energy consumption per image at task level compared to the state-of-the-art (SOTA) SRAM CiM macros.

A. Measurement Setup

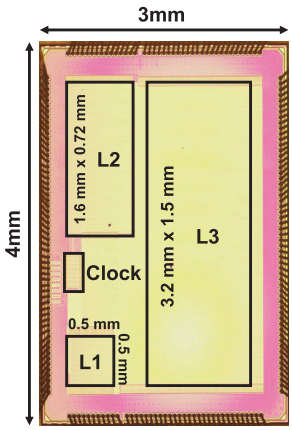
Fig. 12 illustrates the test platform of the fabricated chip, including a chip board, three DC power supply boards, and a main board. The FPGA is stacked on the main board. The FPGA sends the testbench data to the test chip and receives the outputs from the chip. The FPGA is controlled by a computer via an Ethernet connection.

Based on the chip measurement results, a detailed system-level performance breakdown is evaluated through post-layout simulation using Synopsys PrimeTime. Inference accuracy is obtained by end-to-end PyTorch simulation. To improve accuracy, the quantization-aware training (QAT) method is employed during training. The ResNet-56 model is pre-trained on the CIFAR-10 dataset.

B. Performance, Power, and Area Evaluation

Fig. 13 presents the die photograph and chip summary. The 3-Mb fully digital ROM CiM chip was fabricated using a 65nm CMOS technology. This chip includes three DCiROM macros and a clock module. The three macros are supplied by three separate voltage sources. The input/output buffers and test circuits are integrated into the DCiROM macros. The total area of the three macros is 6.205mm^2. The DCiROM chip loads 4-bit signed weights and supports unsigned inputs ranging from 4 to 8 bits. When operating at 0.6V and 1.2V,

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Technology	CMOS 65nm
Die size	3.0×4.0 mm^2
Macro size	6.205 mm^2
Memory size	3024 Kb
Input / Weight / Output precision	4~8-bit / 4-bit / Full
Supply voltage	0.6-1.2 V
Frequency	130-800 MHz
Throughput	3.34-11.23 TOPS
Energy efficiency	38.0-9.0 TOPS/W
Area efficiency	0.55-2.06 TOPS/mm^2

Fig. 13. Die photograph and chip summary.

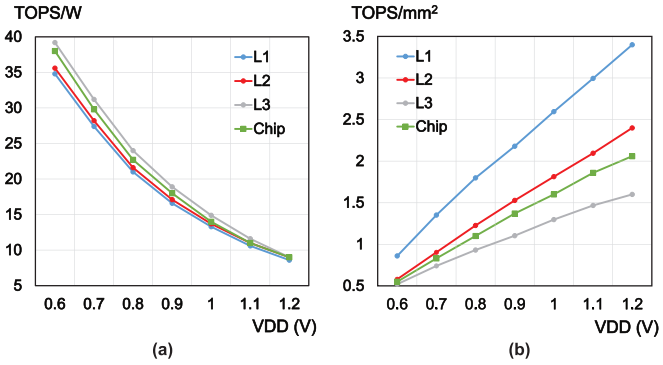


Fig. 14. Chip measurement results: (a) Energy efficiency and (b) Computing efficiency at different supply voltages.

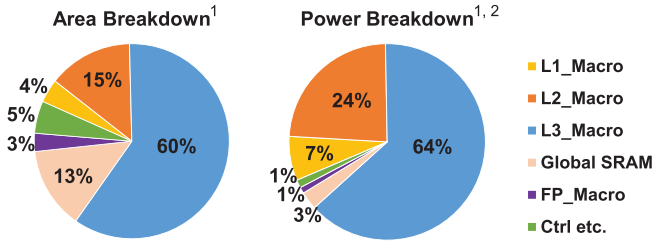
the chip achieves maximum clock frequencies of 130MHz and 800MHz, respectively. The overall chip throughput ranges from 3.34 to 11.23 TOPS.

Fig. 14 illustrates the chip measurement results. L1, L2, and L3 store 144Kb, 576Kb, and 2304Kb of parameters, respectively. Fig. 14(a) shows the energy efficiency of L1, L2, L3, and the entire chip. A standalone L3 achieves the highest energy efficiency of 39.2 TOPS/W at 0.6V. Fig. 14(b) presents the computing density of L1, L2, L3, and the entire chip at different voltages. A standalone L1 achieves a high computing density of 3.4 TOPS/mm^2 at 1.2V.

Fig. 15 shows the area and power breakdown of the entire ResNet-56 end-to-end inference system. The system comprises three DCiROM macros, a floating-point CiM macro, a global SRAM array, and peripheral circuits. The area, energy, and latency data of the components other than DCiROM macros are obtained through post-simulation under a 0.9V supply and a 500MHz clock frequency. The three DCiROM macros account for 88% of the total area and 95% of the total power consumption. In this system, the majority of the total energy and area is utilized for computation.

C. Flexibility Measurement

Fig. 16 illustrates the inference accuracy across different datasets. The baseline is computing using FP32 format.



[1]: Simulated at 0.9V, 1GHz clock, layout level.

[2]: Off-chip and I/O power is not included. Off-chip operation time is not included.

Fig. 15. Area and power breakdown of the proposed design during end-to-end task-level inference.

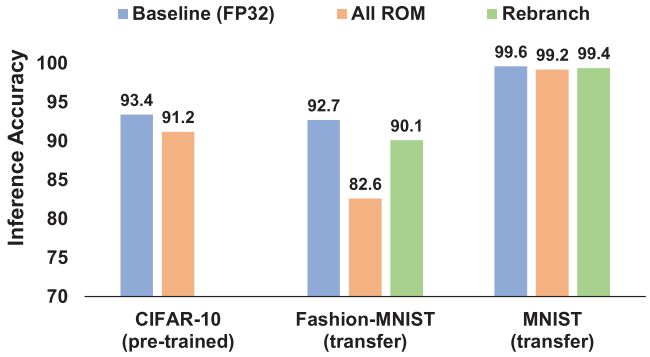


Fig. 16. Inference accuracy of the DCiROM macro in pre-trained and transfer learning datasets.

The parameters of ResNet-56 are pre-trained on the CIFAR-10 dataset. As a result, ReBranch is not required for the CIFAR-10 dataset, and the inference accuracy achieves 91.2%. Compared with the baseline, there is an approximate 2.2% accuracy reduction due to model quantization.

ReBranch architecture is an efficient approach to improving inference accuracy in transfer learning. As shown in Fig. 16, ReBranch enhances accuracy by up to 7.5% compared with all-ROM-based inferences. Through ReBranch transfer learning, the accuracy loss is reduced to 0.2% and 2.6% on the MNIST and Fashion-MNIST datasets, respectively, compared to the FP32 baseline.

D. Metrics Comparison

The comparison is summarized in Table II. As a fully digital CiM chip, this work integrates 3024Kb of on-chip parameters, achieving an ultra-high memory density of 487 Kb/mm^2. It delivers the highest throughput of 12.81 TOPS at a 1.2V voltage supply. Additionally, at a 0.6V voltage supply, it achieves the highest energy efficiency of 38.0 TOPS/W. Leveraging the ROM-logic fusion CiM architecture, this work attains a record-high density figure of merit (FoM) of 1004 TOPS/mm^2· Kb/mm^2 when normalized to 65nm, which is 6.9-29.5x higher than the SRAM/DRAM CiM counterpart. Compared with the analog ROM CiM works [16], [17], the FoM of this chip is 10.2x higher. These results demonstrate the potential of this architecture for deploying large-scale DNNs on-chip.

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TABLE II
COMPARISON TABLE WITH PREVIOUS CIM WORKS

	ISSCC'24 [8]	JSSC'23 [22]	CICC'24 [17]	ISSCC'25 [9]	ISSCC'25 [10]	This work	
Technology	65nm	28nm	28nm	28nm	28nm	**65nm**	
CiM operation	Analog SRAM	Analog eDRAM	Analog ROM	Hybrid SRAM	Digital SRAM	**Digital ROM**	
Capacity (Kb)	80	9,600	22,528	64	32	**3,024**	
Precision (Input/Weight/Output)	4~8/4~8/Full	4,8/5,9/8	1~8/2~8/Full	8/8/Full	4/8/FP8/BF16	**1~8/4/Full**	
Voltage (V)	0.6~1.1	1.1	0.7~1.1	0.6~0.9	0.55~0.9	**0.6**	**1.2**
[2]Throughput (TOPS)	0.38	19.5	0.88~1.6	0.16~0.68	0.50	**3.41**	**12.81**
[2]Energy efficiency (TOPS/W)	256	115.8	80~168	84~271	70~200	**38.0**	**9.0**
[1,2]Energy efficiency (TOPS/W)	256	21.49	14.9~31.2	15.6~50.3	13.0~37.1	**38.0**	**9.0**
[2]Computing density (TOPS/mm^2)	0.78	2.03	0.32~0.64	1.48~6.28	11.56	**0.55**	**2.06**
[1,2]Computing density (TOPS/mm^2)	0.78	0.19	0.030~0.059	0.14~0.58	1.07	**0.55**	**2.06**
Memory density (Kb/mm^2)	167	973	8,928	594	732	**487**	
[1]Memory density (Kb/mm^2)	167	181	1,656	110	136	**487**	
[2,3]FoM (TOPS/mm^2·Kb/mm^2)	130	1,975	2,856~5,713	879~3,730	8,462	**268**	**1,004**
[1,2,3]FoM (TOPS/mm^2·Kb/mm^2)	130	34	49~98	15~64	146	**268**	**1,004 (6.9-29.5x)**

[1]: Normalized to 65nm, assuming energy efficiency ∝ (tech.)^2, area ∝ (tech.)^2, computing density^{-1} ∝ 2 x (tech.)^2.

[2]: 1 OP = 1 4b-4b addition/multiplication, off-chip and IO not included.

[3]: FoM = computing density x memory density.

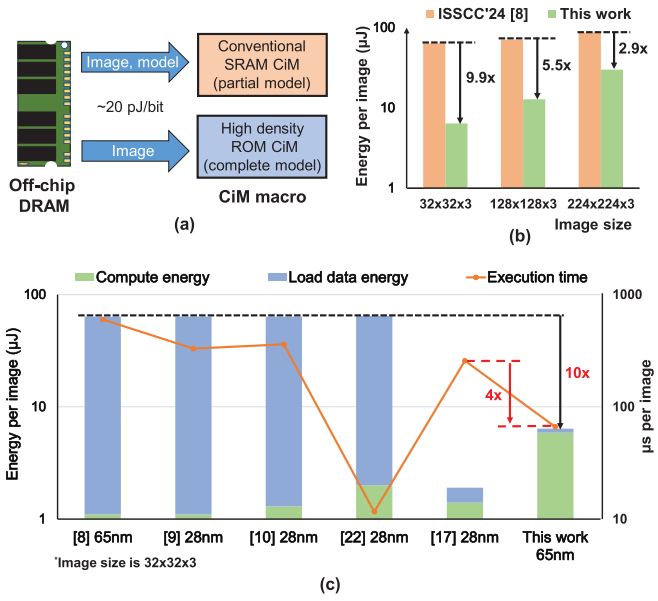


Fig. 17. Task-level measurement evaluations: (a) Off-chip data transport energy, (b) Inference energy comparison with prior SRAM CiM work [8] for different image sizes, (c) Energy consumption and execution time compared with SOTA CiM works.

Fig. 17 evaluates task-level energy consumption during end-to-end inference. In the same technology node, SRAM CiM requires more than 3x the area to load an entire DNN compared with ROM CiM. Generally, SRAM CiM has a relatively small capacity, allowing it to load only a partial model on-chip. As a result, weights must be reloaded from off-chip DRAM during each inference, resulting in significant energy consumption. As shown in Fig. 17(a), the data transport energy is estimated at 20 pJ/bit [32]. In contrast, high-density ROM CiM only needs to load image data from the external buffer. Fig. 17(b) illustrates the energy consumption per inference for different image sizes. Although the SRAM CiM work using an advanced process exhibits better energy efficiency at macro level, this work reduces energy consumption by 2.9-9.9x at task level. Smaller image sizes result in greater energy improvements. Fig. 17(c) shows the ratio of computing energy and the execution time per inference. Compared to previous SRAM/DRAM CiM chips, this work reduces energy consumption by nearly 10x. Additionally, compared with the 28nm analog ROM CiM chip, this work improves inference speed by almost 4x.

V. MORE DISCUSSION

The proposed DCiROM mitigates the challenges of computing density in previous ROM CiM works. Through ROM CiM resource reuse, DCiROM further enhances on-chip memory density. Beyond ResNet-56, the DCiROM design approach can be extended to other lightweight DNNs. While this chip achieves ultra-high capacity, challenges remain in deploying larger models on-chip.

A. Limitation

As the theory of neural networks and the practice of architectures continue to evolve, various new structures are emerging, and the scale of models is increasing continuously. Although DCiROM demonstrates ultra-high memory density, a gap remains in deploying large language models (LLMs) on-chip, such as BERT and MiniGPT-4. The proposed ROM-logic

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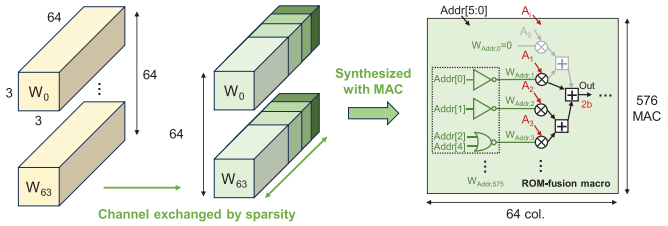


Fig. 18. Data sparsity optimization and weight compression.

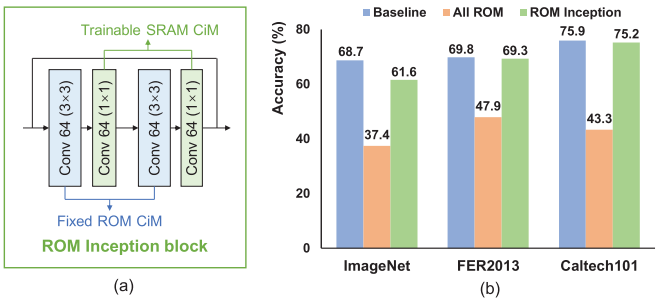


Fig. 19. (a) ROM Inception architecture, and (b) inference accuracy on complex tasks.

fusion CiM has limitations when the memory-compute ratio exceeds 64, which restricts further improvements in on-chip memory density.

The ReBranch architecture provides task-level flexibility within the same domain, such as from ImageNet (item classification) to CIFAR-10 (item classification). However, for transfer learning across different task domains, such as from ImageNet (item classification) to FER2013 (facial expression recognition), ReBranch experiences an accuracy loss of over 40%. This limitation stems from the inability to reconstruct the weights in the fixed trunk ROM CiM.

B. Sparsity Optimization

Data sparsity and weight compression optimization in ROM CiM have not yet been thoroughly explored. Although 8-bit quantization is the most commonly used in practice, quantization with fewer bits, such as 4-bit, holds significant potential because of the possibility of higher data sparsity. Investigating model structures to ensure inference accuracy with fewer quantized bits represents a promising research direction.

DNN models, such as VGGNet and ResNet, often contain a large number of redundant parameters. As a result, weight compression is a potential approach to further improving on-chip capacity. Additionally, row exchange offers a potential method to reduce computations using a pipeline adder tree. As illustrated in Fig. 18, the proposed methodology combines sparsity optimization and weight compression to enhance storage efficiency. Multiple bits can be compressed via a lookup table (LUT)-based scheme. Simulation results demonstrate a more than 25 Mb/mm^2 memory density in a 28nm CMOS process. The DCiROM architecture, enhanced with sparsity-aware compression, presents a scalable solution for deploying larger-scale models. Moreover, chiplet-based design can further extend applicability when weight storage demands exceed single-chip capacity.

C. Flexibility Enhancement

YOLoC and Hidden-ROM architectures have been proposed for analog ROM CiM with stringent array structures. However, previous flexibility architectures require separate circuits for digital-domain ROM CiM, resulting in additional area overhead. A more suitable architecture is needed to achieve flexibility in digital ROM CiM.

Similar to the ReBranch paradigm, transfer learning based on existing network architectures represents an efficient design strategy. Drawing inspiration from MobileNet and Inception architectures, the point-wise convolution as a possible solution for enhancing flexibility in digital ROM CiM systems. As illustrated in Fig. 19(a), a promising approach incorporates trainable pointwise convolution layers between fixed ROM CiM layers to enable adaptive feature transformation.

Fig. 19(b) presents the inference accuracy of the proposed ROM-Inception architecture, which builds upon a sparsity-optimized ResNet-18 model pre-trained on ImageNet. Experimental results demonstrate that this architecture successfully maintains task-level flexibility when processing complex datasets, while preserving the density benefits of ROM CiM.

VI. Conclusion

This paper introduces DCiROM, a novel digital compute-in-ROM design approach for deep neural network inference applications. The proposed approach utilizes a ROM-logic fusion CiM architecture that overcomes the trade-off between the high memory density of ROM and the high computing density of digital computation. Furthermore, the proposed ROM CiM resource reuse method achieves flexibility while minimizing area overhead. The design is implemented in a fabricated chip using a 65nm CMOS technology, achieving a memory density of 487 Kb/mm^2 and a computing density of 2.06 TOPS/mm^2. The chip loads all convolution layers of ResNet-56, significantly reducing off-chip memory access overhead during end-to-end inference. As a result, its inference energy consumption per image is 10x lower than other SRAM CiM works at task level.

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